

Inference for a Single Mean

Estimating a population average

Download the section 15 .Rmd handout to
STAT240/lecture/sect15-single-mean.

Download the file TIM.txt to STAT240/data.

The Boston Marathon is a prestigious 26.2 mile annual race.

It is held in April, but was held in October in 2011.

TIM.txt contains times of Boston Marathon runners from 2010 to 2011.

Possible questions of interest

- Average running time for a given year
- Change in running time from 2010 to 2011

Let's tidy and explore the data first.

What is the average time μ of all finishers? Write each individual observation as a random variable X_i .

We must assume that each X_i is independent with the same distribution with mean μ and sd σ .

$$X_i \sim D(\mu, \sigma)$$

Let's focus on the 3557 18-34 female finishers from 2010. The point estimate for μ is the sample mean

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

By the CLT, this variable is approximately normal for large enough n .

Our observed mean is $\bar{x} = 235.5$. How reliable is this guess?

$E(\bar{X}) = \mu$, $sd(\bar{X}) = \frac{\sigma}{\sqrt{n}}$. The CLT gives

$$\bar{X} \dot{\sim} N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

which will be the basis of our CI and H-test.

A CI, in general, is

point estimate $\pm$ critical value $\times$ standard error

- For μ , the point estimate is $\bar{x}$
- The standard error is $\frac{\sigma}{\sqrt{n}}$

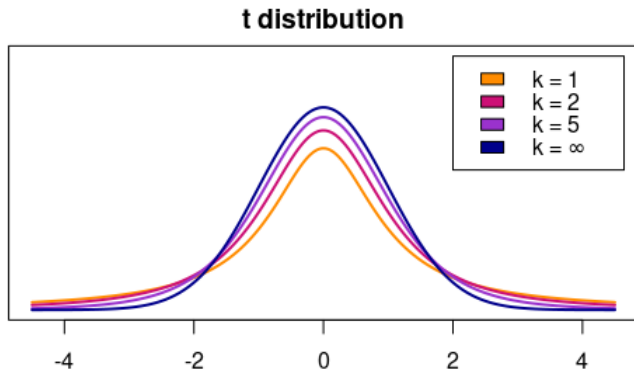
We should be able to use a Z critical value:

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

Problem: we don't have σ . So, use a T value.

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$$

The T bell curve is wider than $N(0, 1)$.



T CI for μ :

$$\bar{X} \pm t_{\alpha/2, n-1} \cdot \frac{S}{\sqrt{n}}$$

We are 95% confident that the true average time for an 18-34 year old female runner in 2010 is within (234.3, 236.7).

Check with t.test.

Is the true average time for an 18-34 year old female runner in 2010 equal to 240 minutes?

$$H_0 : \mu = 240 \quad \text{versus} \quad H_A : \mu \neq 240$$

We need to gather evidence against H_0 with our observed $\bar{x} = 235.5$.

Let's use $\alpha = 0.05$.

If $\mu = 240$, then $\bar{X}$ should be close to 240, and $\bar{X} - 240$ should be small.

We also have to take estimation error (standard error) into account.

$$\frac{\bar{X} - 240}{\sigma/\sqrt{n}}$$

should be close to 0.

Formally, our model is

$$X_i \sim D(\mu, \sigma)$$

and our null is $H_0 : \mu = 240$. The test statistic must have a known distribution if H_0 is true.

$$\frac{\bar{X} - 240}{\sigma/\sqrt{n}} \stackrel{?}{\sim} N(0, 1)$$

will not work.

Instead, use

$$\frac{\bar{X} - 240}{S/\sqrt{n}} \dot{\sim} t_{n-1}$$

Our observed test statistic is -7.41 . For a two-sided p-value, calculate

$$P(t_{n-1} < -7.41) + P(t_{n-1} > 7.41)$$

We get a very small p-value, so we reject H_0 .

We can verify this result in t.test by specifying our null value.

In general, perform a T test on independent draws from the same population. Use hypotheses

$$H_0 : \mu = \mu_0 \quad \text{versus} \quad H_A : \mu \neq \mu_0$$

and test statistic

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$$

and null distribution t_{n-1} .

P-value calculation:

- If the alternative is $H_A : \mu < \mu_0$, the p-value is the area *below* the test statistic.
- If the alternative is $H_A : \mu > \mu_0$, the p-value is the area *above* the test statistic.
- If the alternative is $H_A : \mu \neq \mu_0$, the p-value is $2 \times$ the area *outside* of the test statistic.

Consider female runners age 60-64 from 2010. Do we have evidence that their average time is *less* than 280 minutes?

- Write one-sided hypotheses about μ
- Re-calculate summary statistics
- Perform the T test with $\alpha = 0.02$.
- Check with `t.test`